1 BUILD ONCE

no: fix and retry

Research code
+ its paper

Coding agent

Tool package
- Environment
- { } Functions
- Agent skill
- Tests

```
$ motel onboard <repo>
```

Matches known results?

CO_2 uptake, mmol/g

Tool	1.52
Lab	1.48

2 USE EVERYWHERE

Motel MCP server
serving 50 verified tools

MCP

Any agent harness

Can silicalite capture CO_2 from flue gas?

```
find_tools "CO2 capture"
run_tool zeopp · pores
run_tool raspa3 · adsorption
```

Yes: **18×** more selective for CO_2 than N_2

Verified · Cited · Reproducible

Claude · Codex · Gemini CLI · Cursor · LangGraph

3 THE CATALOG

Quantum chemistry
- PySCF
- Psi4
- Quantum ESPRESSO
- GPAW
- xTB
- CREST

Molecular simulation
- LAMMPS
- OpenMM
- ASE
- ML potentials
- phonopy
- phono3py
- MDAnalysis
- OVITO

Materials
- pymatgen
- Materials Project
- robocrys
- PyXtal
- surfaces
- ion diffusion
- Zeo++
- RASPA3

Thermodynamics
- Cantera
- CoolProp
- pycalphad
- Reaktoro
- thermo
- FiPy

Spectroscopy
- xraylarch
- xraylib
- pyOpenMS
- MixSense NMR
- AtomBridge

Energy
- PyBaMM
- impedance.py
- pvlib

Chemistry & drugs
- RDKit
- RDKit descriptors
- Open Babel
- AiZynthFinder
- AutoDock Vina
- PROPKA
- UMA-Dock
- peptide-agent

Knowledge & lab
- SKY synthesis
- ARIA doping
- MP analogies
- CLUE
- Catalyze
- AtomicShorts